LUCID LIGHT — Outer PCB

Design Overview and Review Notes

Purpose

The PCB is designed to control 6 high-power Bridgelux V22 LEDs in order to produce a light output that could illuminate a quarter acre of land with at least 5 foot-candles.

Each LED will be driven independently at a target current of 2.1 A, using a MAX20078 synchronous buck LED driver circuit.

Power Distribution

Power will enter the PCB through the main power connector at the top of the board and will pass through a fuse rated for 20 A. The fuse's suitability must be confirmed against the conductor ratings and expected fault conditions.

The PCB uses a four-layer design with nominal 2 oz copper on each layer:

- **Layer 1 — Top:** This is where most of the components and electrical routing are located.
- **Layer 2 — GND:** This is a ground plane covering the PCB, except where clearances and cutouts are required.
- **Layer 3 — +48V_Fused:** This is a central fused power-distribution bus supplying the 6 drivers.
- **Layer 4 — Bottom:** This is for additional electrical routing and where the LEDs will be soldered onto the board.

Vias connect the outer-layer traces to the internal power (+48 V) and ground (GND) layers.

The estimated total input current is approximately 9.89 A at 48 V, assuming 33.9 V forward voltage per LED and 90% conversion efficiency for the driver circuits. This is a nominal estimate, not a measured worst-case current.

LED Circuit Operation

Each LED is powered by an independent MAX20078 synchronous buck converter circuit, which steps down the 48 V input to the voltage required by the LED.

The controller operates two external MOSFETs (Q101 and Q102 in the first circuit) to regulate power delivered through the inductor (L101) to the LED.

Current Regulation: The CSP and CSN pins measure the voltage across the current-sense resistor (R105), located in the low-side MOSFET current path. The MAX20078 uses this measurement to regulate the average inductor current, and therefore the LED current.

Current Reference: The REFI pin controls the LED current setpoint. A reference-voltage divider is included only in the first circuit, as the design uses a common reference for all six drivers operating

at the same target current. The resistor values in the REFI reference-voltage divider are determined using the chip datasheet's current-setting relationship and the voltage-divider equation. The common REFI connection should be verified across all six circuits in the final schematic.

Voltage Feedback: The R102/R103 divider provides a scaled output-voltage signal to the OUT pin, which is used by the controller.

Additional Control: The DIM pin provides PWM dimming control, while the bootstrap network supports high-side MOSFET switching. Input and output capacitors provide filtering and voltage stabilization.

All six circuits follow the same basic layout and design, with component numbering grouped from 101 through 601. The three circuits on the right are nearly mirrored versions of the three on the left; however, because the controller's pin arrangement cannot be mirrored, their layouts differ slightly.

Physical PCB Layout

The board currently measures approximately 85 × 170 mm, with a target thickness of 1.6 mm and four nominal 2 oz copper layers. The size may change slightly to accommodate mechanical mounting requirements; any such change will require a layout and clearance review.

The 6 LEDs are arranged in two columns of 3. Each LED has a 24 mm diameter cutout through the PCB, as the LEDs are planned to be mounted on the back with light shining through the openings. The LEDs also have two screw holes to help secure them in place.

The driver circuits are positioned as close to their respective LEDs as practical to minimize routing distance and keep high-current traces short. The main +48 V fused power bus runs down the center of the PCB, with branches supplying the individual circuits.

Additional mounting holes and copper keepouts are included for mechanical attachment.

Review Notes

Please feel free to tear this apart, as this is my first time designing a PCB and I have been learning as I go. There are likely mistakes I have overlooked, and your help is greatly appreciated.